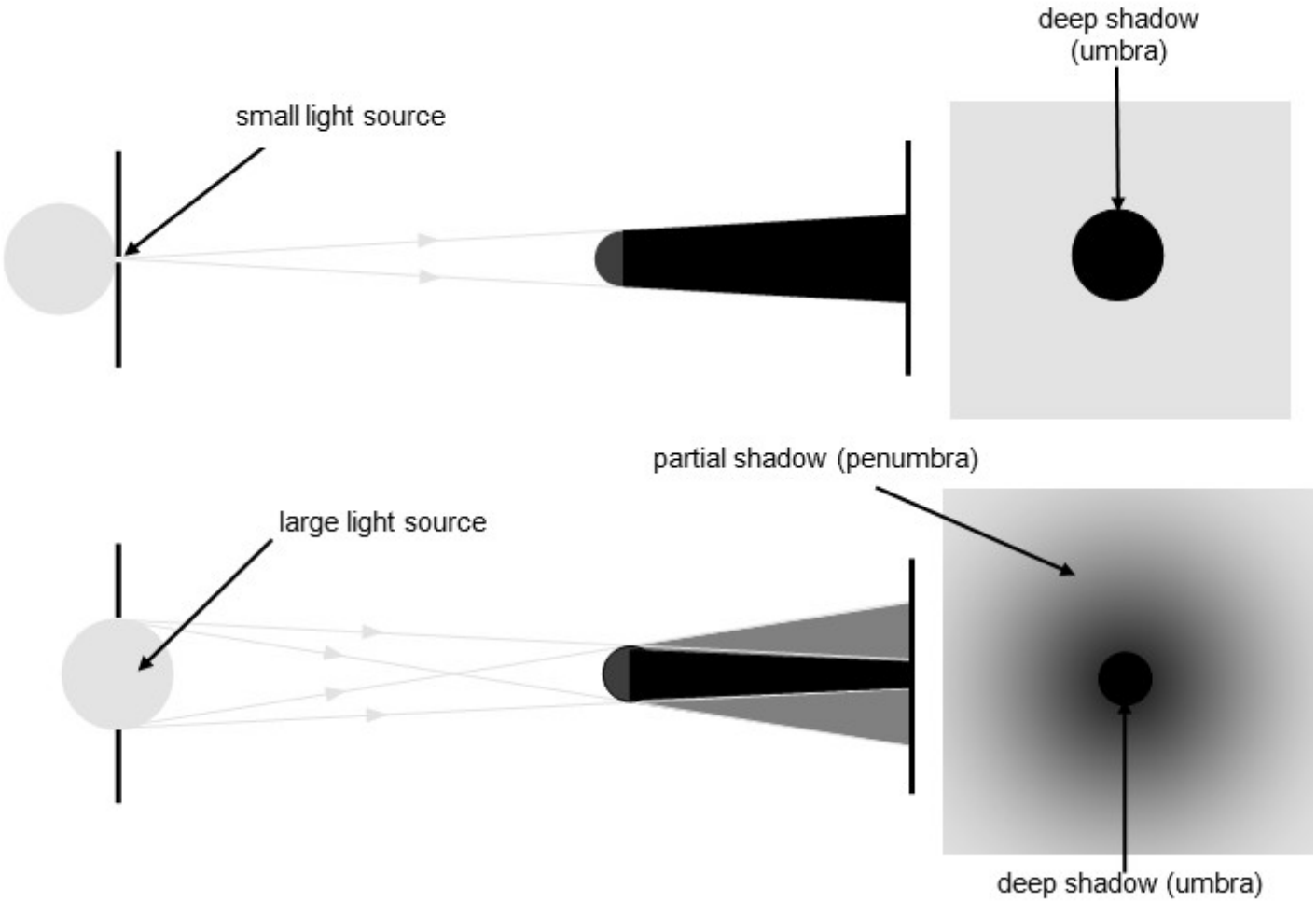
Shadows

Where there is no light there is darkness, we call this a shadow. In fact at night we are in the shadow of the Earth.

You can see from the two diagrams that there are two types of shadow; a crisp edged one formed by a point source of light and a fuzzier one that is formed by a larger source.



The region of deep, total shadow is called the **UMBRA** and the region of partial shadow is called the **PENUMBRA**. The umbra is a region where no light can get to while the penumbra is a region where some light can reach.

Investigation Planning Sheet

1. PLANNING

1.1 Aim

What are you trying to find out? What problem are you trying to solve?

_____

_____

_____

1.2 Variables

You can only change one variable at a time and measure its effect. Select the variable that you would like to investigate. The rest of the variables must be kept the same.

Change (Independent)

_____

Measure (Dependent)

_____

Keep the Same (Controlled)

_____

_____

_____

_____

1.3 Hypothesis

A statement about what you think will happen. This statement needs to link the independent and dependent variables.

_____

_____

_____

_____

2. CONDUCTING AND RECORDING

2.1 Equipment

Write a list of all the equipment you will need to conduct your experiment.

- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____

- _____
- _____

2.2 Procedure

Design a method by which you can slowly change one variable and measure its affect on the other.

1. _____
   _____
2. _____
   _____
3. _____
   _____
4. _____
   _____
5. _____
   _____
6. _____
   _____
7. _____
   _____
8. _____
   _____
9. _____
   _____
10. _____
    _____
    __
11. _____
    _____
    __
12. _____
    _____
    __
13. _____
    _____
    __

14. _____

_____

__

2.3 Results

Construct a table so that you can record the changes in the variables.

3. ANALYSING

3.1 Graphing

Draw a graph of your results. Use the following rules to determine which type of graph to construct. Remember to use a **line of best fit** if you are constructing a line graph.

Independent Data	Dependent Data	Type of Graph
Data is continuous (eg. time)	Data is continuous (eg. height)	Line graph
Data is in categories (eg. eye colour)	Data is numerical (eg. number of people)	Column graph

3.2 Analysing Your Results

Do your results support your hypothesis or not?

_____

_____

Explain **how** you used your results to come to your conclusion above.

_____

_____

_____

_____

4. EVALUATION

4.1 Errors

What are the possible sources of error in your investigation?

_____

_____

_____

How might these sources of error affect your data?

_____

_____

_____

4.2 Improving Your Investigation

What changes would you make to your investigation to improve your results?

_____

_____

_____

_____

_____

5. REPORT

You need to write a report of your experiment. Your report should include the following:

Aim: What is the aim or purpose of your investigation?

Hypothesis: Your prediction in the form of a statement. This needs to link your independent and dependent variables.

Equipment: A list of all the equipment you used in your investigation.

Diagram of Equipment: A labelled diagram of how you set up your equipment to conduct your investigation.

Method: A list of the steps you went through to conduct your investigation.

Results: The results of your investigation presented in both a table and a graph.

Discussion: Comment on whether or not your results supported your hypothesis, what were the possible sources of error in the investigation and what you could do to improve your investigation.

Conclusion: Short statement to sum up your experiment.